14A 1 THE BRØNSTED–LOWRY THEORY

SPECIFICATION REFERENCE
14.1 14.2 14.6

LEARNING OBJECTIVES

- Understand that a Brønsted–Lowry acid is a proton donor and a Brønsted–Lowry base is a proton acceptor and that acid–base reactions involve proton transfer.
- Be able to identify Brønsted–Lowry conjugate acid–base pairs.
- Understand the difference between a strong and a weak acid in terms of the degree of dissociation.

BRØNSTED–LOWRY ACIDS AND BASES

In 1923, physical chemists Johannes Nicolaus Brønsted in Denmark and Thomas Martin Lowry in England independently proposed the theory that carries their names. In the Brønsted–Lowry theory, acids and bases are defined by the way they react with each other.



▲ **fig A** Johannes Nicolaus Brønsted (1879 -1947).



▲ **fig B** Thomas Martin Lowry (1874-1936).

They defined an acid as a substance that can donate a proton, i.e. a **proton donor** (hydrogen ion, H^+). They defined a base as a substance that can accept a proton, i.e. a **proton acceptor**.

We might expect that any substance containing hydrogen could act as an acid. In practice, a substance behaves as an acid only if the hydrogen carries a slight positive charge. As an example, this is the case when it is bonded to a highly electronegative atom to the right of the Periodic Table, e.g. oxygen or a halogen.

In order to accept a proton, a base has to contain a lone pair of electrons that it can use to form a dative covalent bond with the proton. So, a base must contain an atom to the right-hand side of the Periodic Table, and this is often oxygen.

CONJUGATE ACID–BASE PAIRS

An equilibrium is established when hydrogen chloride dissolves in water. The following equation represents this equilibrium:

$$HCl(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + Cl^-(aq)$$

In the forward reaction:

- HCl is acting as an acid because it is donating a proton to H_2O.
- H_2O is behaving as a base as it is accepting a proton from HCl.

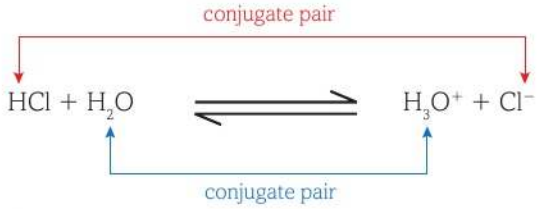
In the reverse reaction:

- H_3O^+ is behaving as an acid because it is donating a proton to Cl^-.
- Cl^- is behaving as a base because it is accepting a proton from H_3O^+.

When the acid HCl loses a proton it forms a base, Cl^-. These two species are called a **conjugate acid–base pair**.

When the acid H_3O^+ loses a proton it forms the base H_2O. These two species also form a conjugate acid–base pair.

So the equilibrium mixture above contains two conjugate acid–base pairs:



▲ **fig C** Conjugate acid–base pairs formed from hydrochloric acid and water.

EXAM HINT

The conjugate base for HCl is the Cl^- ion, which is a very weak conjugate base. However, in the presence of a strong acid (such as concentrated sulfuric acid), even the Cl^- ion will accept a proton to form HCl fumes. You saw this in **Topic 8 (Book 1: IAS)**.

Cl^- is the **conjugate base** of HCl; H_2O is the conjugate base of H_3O^+.

Hydrochloric acid can donate *one* proton – it is called a *monoprotic* or monobasic acid.

Sulfuric acid can donate *two* protons and is therefore called a *diprotic* or dibasic acid.

Step 1: $H_2SO_4 \rightarrow H^+ + HSO_4^-$

Step 2: $HSO_4^- \rightarrow H^+ + SO_4^{2-}$

Similarly, some bases such as the carbonate ion can accept more than one proton:

Step 1: $CO_3^{2-} + H^+ \rightarrow HCO_3^-$

Step 2: $HCO_3^- + H^+ \rightarrow H_2CO_3$

The carbonate ion is therefore described as a *diprotic* or diacidic base.

DID YOU KNOW?

- Acids that donate a maximum of one, two or three protons are called monoprotic, diprotic or triprotic, respectively. They are also called monobasic, dibasic or tribasic.
- Bases that can accept one, two or three protons are called monoprotic, diprotic or triprotic, respectively. They are also called monoacidic, diacidic or triacidic.

We will use the terms monobasic, dibasic, etc. for acids throughout this book.

ANOTHER EXAMPLE OF A CONJUGATE ACID–BASE PAIR

An equilibrium is established when ammonia dissolves in water. The following equation represents this equilibrium:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

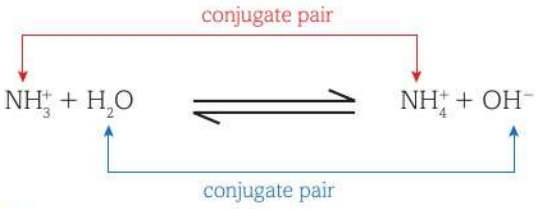
In the forward reaction:

- NH_3 is acting as a base because it is accepting a proton from H_2O.
- H_2O is acting as an acid because it is donating a proton to NH_3.

In the reverse reaction:

- NH_4^+ is behaving as an acid because it is donating a proton to OH^-.
- OH^- is behaving as a base because it is accepting a proton from NH_4^+.

So the equilibrium contains two acid–base conjugate pairs:



▲ **fig D** Conjugate acid–base pairs formed from ammonia and water.

EXAM HINT

The equilibrium that ammonia forms with water helps us to understand the test for the ammonium ion that you saw in **Topic 8 (Book 1: IAS)**. When we add hydroxide ions to a solution containing ammonium ions, the equilibrium shifts to the left. Ammonia gas is liberated and we can test for it using damp red litmus paper.

AMPHOTERIC SUBSTANCES

In its reaction with HCl, H_2O behaves as a base by accepting a proton. However, in its reaction with NH_3, H_2O behaves as an acid by donating a proton.

A substance that can act as either an acid or a base is described as being **amphoteric**.

DID YOU KNOW?

You may also see the term amphiprotic. An amphiprotic substance is one that can both donate and accept protons. So water is described as being amphiprotic. Other examples of amphiprotic substances are amino acids and the hydrogen sulfate ion, HSO_4^-.

All amphiprotic substances are also amphoteric, but the opposite is not true. There are amphoteric substances, like aluminium oxide, that do not donate or accept protons when they act as acids or bases, respectively. These substances are in the category of acting as acids or bases according to the Lewis theory. The Lewis theory states that an acid is an electron pair acceptor and a base is an electron pair donor.

The advantage of the Lewis theory is that it shows the similarity between acid–base reactions that involve proton transfer and other acid–base reactions that do not. It is a useful way of explaining why some reactions in organic chemistry that are catalysed by acids can also be catalysed by some substances that can accept a pair of electrons. For example, the nitration of benzene is catalysed by sulfuric acid, but the reaction also occurs if nitric acid is used with boron trifluoride, BF_3.

Neither the term 'amphiprotic' nor the Lewis theory of acids and bases is required learning for your International A Level course.

NITRIC ACID AS A BASE

The most common method to nitrate benzene (see **Topic 18**) is to use the 'nitrating mixture' of concentrated nitric acid and concentrated sulfuric acid. This mixture forms the equilibrium:

$$H_2SO_4 + HNO_3 \rightleftharpoons HSO_4^- + H_2NO_3^+$$

Acid 1 Base 2 Base 1 Acid 2

In this reaction:

- H_2SO_4 is an acid; its **conjugate base** is HSO_4^-
- HNO_3 is a base; its **conjugate acid** is $H_2NO_3^+$.

It seems strange to refer to nitric acid as a base. However, in this reaction the nitric acid *is* behaving as a base.

STRONG AND WEAK ACIDS

A strong acid is defined as one that is almost completely **dissociated** in aqueous solution.

Virtually all the hydrogen chloride molecules are dissociated in a dilute solution of hydrochloric acid. We usually represent this by using a single direction arrow in the equation for the dissociation:

$$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$$

LEARNING TIP

From now on, we are going to use $H^+(aq)$ to represent the acid protons in aqueous solution, rather than H_3O^+. It is not certain how many water molecules are bonded to a given hydrogen ion. The formula could just as easily be $H_5O_2^+$ or $H_7O_3^+$. For this reason, $H^+(aq)$ is possibly the best representation of acid protons in aqueous solution. The choice of formula should not affect your understanding of the concepts involved.

LEARNING TIP

Ionisation is also used to describe the conversion of acid molecules into ions when an acid dissolves in water. You may see this term used in other books.

DID YOU KNOW?

Although, strictly speaking, an equilibrium exists between HCl molecules, H^+ ions and Cl^- ions, the acid is considered to be completely dissociated in dilute solution. So, strong acids are often said to be 100% dissociated.

By contrast, a weak acid is defined as one that is only **partially dissociated** (often less than 10%) in aqueous solution. Organic acids such as ethanoic acid are typically weak acids.

We represent partial dissociation by using the reversible arrow sign ($\rightleftharpoons$) in the equation for the dissociation:

$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$

CHECKPOINT

1. Give the formula of the conjugate acid of each of the following species:
   (a) CH_3COO^-
   (b) CH_3NH_2
   (c) HSO_4^-
2. Give the formula of the conjugate base of each of the following species:
   (a) $HClO_4$
   (b) H_3O^+
   (c) HSO_4^-
3. In each of the following examples, identify the two conjugate acid–base pairs. In each case, identify the species that are acting as a Brønsted–Lowry acid.
   (a) $H_2CO_3 + H_2O \rightleftharpoons HCO_3^- + H_3O^+$
   (b) $HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+$
   (c) $CH_3COOH + HNO_3 \rightleftharpoons CH_3COOH_2^+ + NO_3^-$
4. Explain why the following reaction may be described as an acid–base reaction:
   $NH_4^+ + NH_2^- \rightarrow 2NH_3$

SUBJECT VOCABULARY

proton donor an acid is a proton donor
proton acceptor a base is a proton acceptor
conjugate acid–base pair either a base and its conjugate acid or an acid and its conjugate base
conjugate base when an acid donates a proton, the species formed is the conjugate base of the acid
conjugate acid when a base accepts a proton, the species formed is the conjugate acid of the base
amphoteric (substance) a substance that can act both as an acid and as a base
dissociated acid molecules are said to be dissociated when they have split to form ions
partially dissociated only a small fraction of the acid molecules have dissociated

14A 2 HYDROGEN ION CONCENTRATION AND THE pH SCALE

SPECIFICATION REFERENCE
14.3 14.4 14.5 14.6 14.7 14.8 14.9 14.12 PART

LEARNING OBJECTIVES

- Be able to define the term 'pH'.
- Be able to calculate pH from the hydrogen ion concentration.
- Be able to calculate the concentration of hydrogen ions in a solution, in mol dm^{-3}, from its pH, using the expression $[H^+] = 10^{-pH}$.
- Understand the difference between a strong acid and a weak acid in terms of the degree of dissociation.
- Be able to calculate the pH of a strong acid.
- Be able to deduce the expression for the acid dissociation constant, K_a, for a weak acid.
- Be able to calculate the pH of a weak acid from K_a or pK_a values, making relevant assumptions.
- Be able to define the term 'pK_a'.

HYDROGEN ION CONCENTRATION AND pH

STRONG ACIDS

As already mentioned, we assume strong acids are dissociated completely when they are dissolved in water. This means that the hydrogen ion concentration is related directly to the concentration of the acid.

For example, a solution of HCl of concentration 0.100 mol dm^{-3} will produce a hydrogen ion concentration of 0.100 mol dm^{-3}.

The **pH** of an aqueous solution is related to the hydrogen ion concentration by the following equation:

$$pH = -\lg [H^+] \quad \text{or} \quad pH = \lg \frac{1}{[H^+]}$$

The hydrogen ion concentration, $[H^+]$, is measured in mol dm^{-3}.

Which equation you decide to use is a matter of personal preference.

DID YOU KNOW?

Logarithms can only be taken of a number, not a quantity with a unit. So, strictly speaking, the hydrogen ion concentration has to be divided by the standard concentration, $c^\ominus$, which has a value of 1 mol dm^{-3}.

Therefore, the correct expression for calculating pH is $-\lg [H^+]/c^\ominus$. This, however, is not something you need to consider at this level.

LEARNING TIP

Do not worry if you are unfamiliar with the use of logarithms. They are merely a way to convert a scale of numbers in powers of 10 to a linear scale. For example:

lg 100 = 2 (as 100 is 10^2)

lg 10 = 1 (as 10 is 10^1)

lg 1 = 0 (as 1 is 10^0)

lg 0.01 = −2 (as 0.01 is 10^{-2})

The lg of a number can easily be found by using the 'lg' or 'log' button on your calculator.

The accepted abbreviation for logarithm to the base 10 is lg, but you can also use $\log_{10}$ or even log.

WORKED EXAMPLE 1

1 Calculate the pH of aqueous solutions of the following monobasic strong acids. In each case, assume that the acid is completely dissociated. Give your answers to two decimal places.

(a) 0.00100 mol dm^{-3} HCl

(b) 0.0500 mol dm^{-3} HNO_3

(c) 0.150 mol dm^{-3} HBr

Answer

(a) $pH = -\lg [H^+]$
$= -\lg (0.00100)$
$\lg (0.00100) = -3$
so, $-\lg (0.00100) = +3.00$
$pH = 3.00$

(b) $pH = -\lg (0.0500)$
$\lg (0.0500) = -1.30$
$pH = 1.30$

(c) $pH = -\lg (0.150)$
$\lg (0.150) = -0.82$
$pH = 0.82$

2 Calculate the pH of an aqueous solution of 10.00 mol dm^{-3} of HCl. In this solution the HCl is 55% dissociated. Give your answer to two decimal places.

Answer

$[H^+] = 0.55 \times 10.00$ mol $dm^{-3} = 5.50$ mol dm^{-3}
$pH = -\lg (5.50)$
$\lg (5.50) = 0.74$
so, $pH = -0.74$

DID YOU KNOW?

pH and activity

The measured pH of a concentrated acid is never as low as the calculated value, even though the acid may be completely dissociated. This is because the ions that are close together in solution interact with one another. This makes their effective concentration less than the actual concentration. This effective concentration is called the activity. For example, a solution of HCl of 1.00 mol dm^{-3} has an effective hydrogen ion concentration of 0.81 mol dm^{-3}. This makes the pH of this solution 0.09, not 0.00, as calculated from a hydrogen ion concentration of 1.00 mol dm^{-3}.

Only at concentrations below 0.10 mol dm^{-3} do activities and concentrations have similar values.

This activity effect means that the minimum pH and maximum pH that concentrated solutions can have are 0.30 and 14.30, respectively. This activity effect can be ignored at International A Level, which explains the calculation of –0.74 above.

CALCULATING HYDROGEN ION CONCENTRATION FROM pH

It is a simple matter to calculate the hydrogen ion concentration from a given pH value.

You should use the following equation:

$$[H^+(aq)] = 10^{-pH}$$

WORKED EXAMPLE 2

Calculate the hydrogen ion concentration of a solution with a pH of 4.8:

$[H^+(aq)] = 10^{-4.8}$

$= 1.58 \times 10^{-5}$ mol dm^{-3}

To solve this calculation, press the 10^X or $10^{\blacksquare}$ button on your calculator. Then enter the negative pH value, followed by the equals button. The answer will then be displayed.

WEAK ACIDS

Determining the hydrogen ion concentration of an aqueous solution of a weak acid is more complicated. This is because a significant amount of undissociated acid is present in solution. It is necessary to refer to the acid dissociation constant, K_a, for the acid.

ACID DISSOCIATION CONSTANT

If we use HA to represent a weak acid, then the equation for its dissociation in aqueous solution is:

$$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$$

When we apply the equilibrium law to this reaction we obtain:

$$\frac{[H^+(aq)][A^-(aq)]}{[HA(aq)]} = \text{a constant}$$

This constant is called the acid dissociation constant and is given the symbol K_a.

We will now calculate the hydrogen ion concentration of an aqueous solution of ethanoic acid of concentration 0.0500 mol dm^{-3}. The value of K_a for ethanoic acid is 1.74×10^{-5} mol dm^{-3} at 298 K.

$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$

$$K_a = \frac{[CH_3COO^-(aq)][H^+(aq)]}{[CH_3COOH(aq)]}$$

Every time a molecule of CH_3COOH dissociates, a CH_3COO^- ion and a H^+ ion are formed.

This means that $[CH_3COO^-(aq)] = [H^+(aq)]$.

So, the expression for K_a can be simplified to:

$$K_a = \frac{[H^+(aq)]^2}{[CH_3COOH(aq)]}$$

At this stage, it is important to recognise that the concentrations in the expression for K_a are the *equilibrium* concentrations. However, if the value of K_a is very small (as it is in this case) then the concentration of the undissociated acid at equilibrium is very similar to the initial concentration of the acid. It is therefore reasonable to say the concentration at equilibrium is the same as the initial concentration.

Therefore,

$$K_a = \frac{[H^+(aq)]^2}{0.0500} = 1.74 \times 10^{-5} \text{ mol dm}^{-3}$$

$$[H^+(aq)] = \sqrt{(0.0500 \times 1.74 \times 10^{-5})} = 9.33 \times 10^{-4} \text{ mol dm}^{-3}$$

This gives a pH of 3.03.

LEARNING TIP

If you are worried that the approximation we made in the calculation is not justified, then it is possible to solve the precise equation:

$K_a = [H^+(aq)]^2/(0.0500 - [H^+(aq)]) = 1.74 \times 10^{-5}$ mol dm^{-3}

The solution to this equation gives a value for $[H^+(aq)]$ of 9.327×10^{-4} mol dm^{-3}

The difference is significant only for very accurate work.

There is also one other approximation that we made in both the original calculation and this calculation. As we will see in **Section 14A.3**, water ionises very slightly to form hydrogen ions and hydroxide ions. We have ignored the contribution from the dissociation of water to the total hydrogen ion concentration. However, unless the acid is *very* dilute, this approximation is also justified.

The following equation can be used to calculate the hydrogen ion concentration of an aqueous solution of a weak monobasic acid:

$$[H^+(aq)] = \sqrt{(K_a \times [acid])}$$

where K_a is the dissociation constant for the weak acid.

K_a AND pK_a VALUES

Table A (next page) lists some weak monobasic organic acids together with their K_a and **pK_a** values at 298 K, where:

$$pK_a = -\lg K_a$$

You will notice that the *larger* the value of K_a the stronger the acid.

By contrast, the *smaller* the value of pK_a the stronger the acid.

NAME OF ACID	FORMULA OF ACID	K_a/mol dm^{-3}	pK_a
propanoic acid	CH_3CH_2COOH	1.35×10^{-5}	4.87
ethanoic acid	CH_3COOH	1.74×10^{-5}	4.76
benzoic acid	C_6H_5COOH	6.31×10^{-5}	4.20
methanoic acid	$HCOOH$	1.60×10^{-4}	3.80
chloroethanoic acid	$CH_2ClCOOH$	1.38×10^{-3}	2.86
dichloroethanoic acid	$CHCl_2COOH$	5.13×10^{-2}	1.29
trichloroethanoic acid	CCl_3COOH	2.24×10^{-1}	0.65

increasing acid strength

table A

CALCULATING THE pH OF A DIBASIC ACID

Sulfuric acid is the most common dibasic acid. It dissociates in two stages:

$$H_2SO_4(aq) \rightarrow H^+(aq) + HSO_4^-(aq)$$

$$HSO_4^-(aq) \rightleftharpoons H^+(aq) + SO_4^{2-}(aq)$$

H_2SO_4 is a strong acid and is therefore fully dissociated. HSO_4^- is a weak acid ($K_a = 0.0100$ mol dm^{-3}.)

In a 0.500 mol dm^{-3} aqueous solution of H_2SO_4 the contribution to the [H^+(aq)] from the H_2SO_4 will be 0.500 mol dm^{-3}. If we assume that the contribution to the [H^+(aq)] from the HSO_4^- ion is x mol dm^{-3}, then we have the following relationship:

$$K_a(HSO_4^-) = 0.0100 = (0.500 - x)x/0.500$$

Solving this quadratic equation gives $x = 0.0098$.

This gives a total [H^+(aq)] of 0.05098 mol dm^{-3} with a subsequent pH of 0.293.

This calculation shows that the contribution to the [H^+(aq)] from the HSO_4^- ions is negligible. This is because its dissociation is significantly reduced because of the high [H^+(aq)] from the full first ionisation of the H_2SO_4.

A second interesting reason for performing this calculation is because many books state that 0.5 mol dm^{-3} H_2SO_4(aq) can be used as the acid solution in a standard hydrogen electrode (see **Topic 16** on standard electrode potentials). As the standard hydrogen electrode requires a [H^+(aq)] of 1.00 mol dm^{-3} (pH = 0.00), this is clearly incorrect.

CHECKPOINT

1. Calculate the pH of each of the following aqueous solutions of strong monobasic acids. Assume the acid is fully dissociated in each case. Give your answers to two decimal places.
   (a) 0.0100 mol dm^{-3} HI
   (b) 0.500 mol dm^{-3} HNO_3
   (c) 0.00405 mol dm^{-3} HCl
2. Calculate the pH of a mixture of 20.0 cm^3 of 1.00 mol dm^{-3} HCl(aq) and 5 cm^3 of 1.00 mol dm^{-3} NaOH(aq).
3. Calculate the pH of each of the following aqueous solutions of weak monobasic acids. Give your answers to two decimal places.
   (a) 0.100 mol dm^{-3} HCOOH [K_a(HCOOH) = 1.60×10^{-4} mol dm^{-3}]
   (b) 1.00 mol dm^{-3} HF [K_a(HF) = 5.62×10^{-4} mol dm^{-3}]
   (c) 0.505 mol dm^{-3} NH_4Cl [$K_a(NH_4^+) = 5.62 \times 10^{-10}$ mol dm^{-3}]
4. The pH of an aqueous solution of a weak acid, HA, of concentration 0.0305 mol dm^{-3} is 4.97. Calculate the dissociation constant, K_a, for this weak acid.

SKILLS PROBLEM-SOLVING

SUBJECT VOCABULARY

pH (of an aqueous solution) the reciprocal of the logarithm to the base 10 of the hydrogen ion concentration, measured in moles per cubic decimetre, pH = –lg [H^+]. This definition is difficult to remember, so either of the two equations given on page 83 can be used to define pH

pK_a = –lg K_a

14A 3 IONIC PRODUCT OF WATER, K_w

SPECIFICATION REFERENCE
14.10 14.11 14.12 PART

LEARNING OBJECTIVES

- Be able to define the ionic product of water, K_w.
- Be able to define the term 'pK_w'.
- Be able to calculate the pH of a strong base from its concentration, using K_w or pK_w .

DISSOCIATION OF WATER

Pure water has a slight electrical conductivity, so it must contain some ions. It self-ionises according to the following equation:

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

If we apply the equilibrium law to this reaction we obtain:

$$\frac{[H^+(aq)][OH^-(aq)]}{[H_2O(l)]} = \text{a constant}$$

As $[H_2O(l)]$ is constant at a given temperature, the expression may be simplified to:

$$[H^+(aq)][OH^-(aq)] = \text{a constant}$$

This constant is called the **ionic product of water** and is given the symbol K_w.

The value of K_w at 298 K is 1.00×10^{-14} mol^2 dm^{-6}.

A neutral solution is defined as one in which the hydrogen ion concentration is equal to the hydroxide ion concentration. This is the case for pure water.

At 298 K, $[H^+(aq)] = 1.00 \times 10^{-7}$ mol dm^{-3} $[(1.00 \times 10^{-14})^{1/2}]$

So, the pH of pure water at 298 K is 7.00 $[-\lg (1.00 \times 10^{-7})]$

K_w AND pK_w

The relationship between K_w and pK_w is given by the following equation:

$$pK_w = -\lg K_w$$

At 298 K, when $K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6}, $pK_w = 14.00$

Table A gives the values of K_w and pK_w at various temperatures.

TEMPERATURE/K	273	283	293	303	313
K_w/mol^2 dm^{-6}	1.14×10^{-15}	2.93×10^{-15}	6.81×10^{-15}	1.47×10^{-14}	2.92×10^{-14}
pK_w	14.94	14.53	14.17	13.83	13.53

table A

pH OF AQUEOUS SOLUTIONS OF STRONG BASES

An acid dissolved in water produces so many hydrogen ions that the small contribution from the water is insignificant, unless the acid concentration is very small.

However, even the most alkaline solutions contain some hydrogen ions because water ionises.

Sodium hydroxide is a strong base, so in dilute aqueous solutions we can consider its ions to be completely dissociated.

LEARNING TIP

The pH of a neutral solution is often said to be 7.00. However, this is true only for a solution that has a temperature of 298 K.

As with all equilibrium constants, K_w varies with temperature.

At 288 K it has a value of 4.52×10^{-15} mol^2 dm^{-6}.

At this temperature, $[H^+(aq)] = 6.72 \times 10^{-8}$ mol dm^{-3}, giving a pH of 7.17.

So, at 288 K a neutral solution has a pH of 7.17.

Similarly it can be shown that at 308 K, the pH of a neutral solution is 6.84.

LEARNING TIP

The equation should strictly be $pK_w = -\lg K_w/(c^\ominus)^2$, where $c^\ominus$ is the standard concentration of 1 mol dm^{-3}, in order to be able to take the logarithm of a dimensionless quantity. However we will use the simplified version of the equation.

A sodium hydroxide solution of concentration 0.100 mol dm^{-3} therefore has a hydroxide ion concentration of 0.100 mol dm^{-3}.

If $[OH^-(aq)] = 0.100$ mol dm^{-3} and $[H^+(aq)][OH^-(aq)] = 1.00 \times 10^{-14}$ mol^2 dm^{-6}, then

$[H^+(aq)] = 1.00 \times 10^{-14} / 0.100$ mol dm^{-3}

$= 1.00 \times 10^{-13}$ mol dm^{-3}

The pH of this solution is therefore 13.0.

WORKED EXAMPLE

Calculate the pH, at 298 K, of an aqueous solution of potassium hydroxide of concentration 0.0200 mol dm^{-3}.

K_w (298 K) = 1.00×10^{-14} mol^2 dm^{-6}

Answer

As potassium hydroxide is a strong base we may assume that its ions are completely dissociated.

$[OH^-(aq)] = 0.0200$ mol dm^{-3}

So, $[H^+(aq)] = 1.00 \times 10^{-14} / 0.0200 = 5.00 \times 10^{-13}$ mol dm^{-3}

$pH = -\lg (5.00 \times 10^{-13}) = 12.3$

LEARNING TIP

You can use the logarithmic form of the equation $[H^+(aq)][OH^-(aq)] = 1.00 \times 10^{-14}$, which is pH + pOH = 14.0, where $pOH = -\lg [OH^-(aq)]$ if you do not like working with the logarithms of small numbers.

So, in the worked example above:

pH = 14.0 – pOH

= 14.0 – (–lg (0.0200))

= (14.0 – 1.70) = 12.3

CHECKPOINT

1. The ionic product of water, K_w, has a value of 1.00×10^{-14} mol^2 dm^{-6} at 298 K and a value of 6.81×10^{-15} mol^2 dm^{-6} at 293 K. Use this information, where relevant, to answer the following questions.

(a) Calculate the pH of water at (i) 298 K and (ii) 293 K.

(b) Even though pure water at 298 K and at 293 K has different pH values, both samples of water are said to be neutral. Explain why.

(c) Is the following reaction exothermic or endothermic?

$H_2O(l) \rightarrow H^+(aq) + OH^-(aq)$

Explain how you arrived at your answer.

2. Calculate the pH at 298 K of each of the following aqueous solutions of strong bases. Assume the base is fully dissociated in each case. Give your answers to two decimal places.

(a) 0.0100 mol dm^{-3} NaOH

(b) 0.0500 mol dm^{-3} $Ca(OH)_2$

(c) 0.0315 mol dm^{-3} KOH

[$K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6} at 298 K]

SKILLS PROBLEM-SOLVING

SUBJECT VOCABULARY

ionic product of water, K_w the product of the concentration of the hydrogen ions and the hydroxide ions, both measured in mol dm^{-3}

$K_w = [H^+(aq)][OH^-(aq)]$

14A 4 ANALYSING DATA FROM pH MEASUREMENTS

SPECIFICATION REFERENCE
14.13(i) 14.13(ii) 14.14 CP11

LEARNING OBJECTIVES

- Be able to analyse data from the following experiments:
  (i) measuring the pH of a variety of substances, including equimolar solutions of strong and weak acids, strong and weak bases, and salts
  (ii) comparing the pH of a strong and weak acid after dilution 10, 100 and 1000 times.
- Be able to calculate K_a for a weak acid from experimental data given the pH of an aqueous solution containing a known mass of acid.

COMPARING SOLUTIONS THROUGH pH MEASUREMENT

STRONG AND WEAK ACIDS

The relative strengths of different acids can be determined by measuring the pH of equimolar aqueous solutions of the acids, at the same temperature.

Table A shows the pH of 0.100 mol dm^{-3} aqueous solutions of various acids at 298 K.

FORMULA OF ACID	HCl	$CHCl_2COOH$	$CH_2ClCOOH$	HCOOH	CH_3COOH	CH_3CH_2COOH
pH OF 0.100 mol dm^{-3} AQUEOUS SOLUTION	1.00	1.14	1.93	2.38	2.87	2.93

decreasing acid strength

table A

The higher the value of the pH, the weaker the acid.

STRONG AND WEAK BASES

The same method can be used to determine the relative strengths of bases.

Table B shows the pH of 0.100 mol dm^{-3} aqueous solutions of various bases at 298 K.

FORMULA OF BASE	NH_3	CH_3NH_2	$(CH_3)_2NH$	$CH_3CH_2NH_2$	$CH_3CH_2CH_2NH_2$	NaOH
pH OF 0.100 mol dm^{-3} AQUEOUS SOLUTION	11.13	11.82	11.83	11.84	11.86	13.00

increasing base strength

table B

The higher the value of the pH, the stronger the base.

SALTS

Table C shows the pH of aqueous solutions of various salts of concentration 0.100 mol dm^{-3} at 298 K.

FORMULA OF SALT	NaCl	KNO_3	CH_3COONa	NH_4Cl	CH_3COONH_4
pH OF AQUEOUS SOLUTION	7.00	7.00	8.88	5.13	7.00

table C

The pH of NaCl is 7.00 because the salt is made from a strong acid (HCl) and a strong base (NaOH).

The same is true for KNO_3, which is a product of the strong acid HNO_3 and the strong base KOH.

An aqueous solution of CH_3COONa is alkaline because it is the product of a weak acid (CH_3COOH) and a strong base (NaOH).

An aqueous solution of NH_4Cl is acidic because it is the product of a strong acid (HCl) and a weak base (NH_3).

EXAM HINT

You can explain why the pH of 0.100 mol dm^{-3} NH_4Cl is acidic by showing how the NH_4^+ ion behaves in aqueous solution. The NH_4^+ ion behaves as a weak acid in aqueous solution, and it dissociates as shown by the following equation:

$NH_4^+ \rightleftharpoons NH_3 + H^+$

An aqueous solution of CH_3COONH_4 is neutral (pH = 7.00) because it is the product of a weak acid (CH_3COOH) and a weak base (NH_3), and the relative strengths of the acid and base are the same. This is shown by their dissociation constants:

CH_3COOH: $K_a = 1.74 \times 10^{-5}$ mol dm^{-3}

NH_3: $K_b = 1.74 \times 10^{-5}$ mol dm^{-3}

LEARNING TIP

K_a is a measure of acid strength and K_b is a measure of base strength.

K_b for a base is calculated in a similar way to K_a for an acid. The equilibrium set up when ammonia dissolves in water is:

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$$

You will not be asked to calculate values for K_b in the exam.

SUMMARY TABLE FOR AQUEOUS SOLUTIONS AT 298 K

SALT OF A STRONG ACID AND A STRONG BASE	pH = 7 (solution is neutral)
SALT OF A WEAK ACID AND A STRONG BASE	pH > 7 (solution is alkaline)
SALT OF A STRONG ACID AND A WEAK BASE	pH < 7 (solution is acidic)
SALT OF A WEAK ACID AND A WEAK BASE	pH depends on relative strengths of acid and base; if $K_a = K_b$, pH = 7 if $K_a > K_b$, pH < 7 if $K_a < K_b$, pH > 7

table D

DID YOU KNOW?

The reason why some salts form alkaline or acidic solutions is that they undergo hydrolysis.

To learn more about this, research 'salt hydrolysis'.

It is important to recognise that different salts have different pH values because this will explain why, for example, the pH of the solution formed when a strong acid reacts with an equivalent amount of a weak base is less than 7. We will meet this concept when we look at acid–base titration curves in **Section 14B.1.**

EFFECT OF DILUTION ON THE pH OF AQUEOUS SOLUTIONS OF ACIDS

STRONG ACIDS

Table E shows the pH of five aqueous solutions of hydrochloric acid. In each case, the acid has been diluted by a factor of ten from 1.00 (1.00×10^0) to 0.000100 (1.00×10^{-4}) mol dm^{-3}. All solutions are at a temperature of 298 K.

CONCENTRATION/mol dm^{-3}	1.00 (1.00×10^0)	0.100 (1.00×10^{-1})	0.0100 (1.00×10^{-2})	0.00100 (1.00×10^{-3})	0.00100 (1.00×10^{-4})
pH	0.00	1.00	2.00	3.00	4.00

table E

You will notice that the pH increases by a factor of one unit for each 10-fold decrease in concentration.

If we follow this to its logical conclusion, then an aqueous solution of hydrochloric acid of concentration 1.00×10^{-8} mol dm^{-3} should have a pH of 8.00. This is clearly nonsense because this would mean that a solution of an acid was alkaline.

Earlier in this topic we mentioned the fact that the contribution to the hydrogen ion concentration from the dissociation of water can usually be ignored. When solutions are as dilute as 10^{-8} mol dm^{-3} this is no longer the case. The pH of 10^{-8} mol dm^{-3} hydrochloric acid is very close to 7, as the contribution to the concentration of hydrogen ions from the water (10^{-7} mol dm^{-3}) is now greater than that of the acid.

WEAK ACIDS

Table F shows the pH of five aqueous solutions of ethanoic acid (CH_3COOH). In each case the acid has been diluted by a factor of ten from 1.00 (1.00×10^0) to 0.000100 (1.00×10^{-4}) mol dm^{-3}. Once again, all solutions are at a temperature of 298 K.

CONCENTRATION/mol dm^{-3}	1.00 (1.00×10^0)	0.100 (1.00×10^{-1})	0.0100 (1.00×10^{-2})	0.00100 (1.00×10^{-3})	0.000100 (1.00×10^{-4})
pH	2.38	2.88	3.38	3.88	4.38

table F

With a weak acid, the pH value increases by a factor of about 0.50 for each 10-fold decrease in concentration.

PRACTICAL SKILLS CP11

Determining K_a of a weak acid from experimental data

Note that this is a different way of doing **Core Practical 11: Finding the K_a value for a weak acid** to that mentioned in your Lab Book. The following experiment can be performed to determine K_a of benzoic acid (C_6H_5COOH), which is a weak monobasic acid.

- Accurately weigh between 0.40 and 0.50 g of benzoic acid. Then dissolve it in a small volume (say 50 cm^3) of deionised water contained in a beaker. (Benzoic acid is not very soluble, so it may be necessary to warm the water to get it to dissolve. Allow the solution to cool before performing the next stage.)
- Transfer the solution to a 250 cm^3 volumetric flask. Add several washings from the beaker using deionised water, and then make up to the mark with deionised water.
- Put a stopper in the flask and then turn the flask upside down several times to mix the solution.
- Withdraw a sample of the solution and place it in a small beaker.
- Measure the pH of the solution using a calibrated pH meter.

Sample results

Mass of benzoic acid = 0.49 g

pH of solution = 3.00

Analysis of results

$C_6H_5COOH(aq) \rightleftharpoons C_6H_5COO^-(aq) + H^+(aq)$

Molar mass of benzoic acid (C_6H_5COOH) = 122 g mol^{-1}

$n(C_6H_5COOH)$ in 250 cm^3 of solution = 0.49/122 mol

So, $[C_6H_5COOH(aq)] = (0.49/122) \times 4$ mol dm^{-3}

pH = 3.00

So, $[H^+(aq)] = 10^{-3.00} = [C_6H_5COO^-(aq)]$

$K_a = (10^{-3.00})^2 \div ((0.49/122) \times 4)$

$= 6.22 \times 10^{-5}$ mol dm^{-3}

The Data Book value is 6.32×10^{-5} mol dm^{-3} at 298 K.

Safety Note: Wear eye protection. Avoid skin contact with, and do not raise dust from, the benzoic acid.

Our experimental result for the K_a of benzoic acid is close to the accepted value. However, that does not necessarily mean that we have performed an accurate experiment. There may have been a number of errors that have, by chance, cancelled out one another.

To start with, we have assumed that the $[C_6H_5COOH(aq)]$ at equilibrium is identical to the original concentration of benzoic acid. Obviously, this is not correct because it must have dissociated slightly to produce a solution of pH = 3.00.

The major uncertainty in this experiment is the measurement of the pH value. Even a small error leads to a large discrepancy in the final answer. A pH of 3.10 would give a final answer of 3.92×10^{-5}, whereas a value of 2.90 would give 9.87×10^{-5} for the value of K_a.

Transfer errors may also be significant when weighing out small amounts.

CHECKPOINT

1. Predict whether aqueous solutions of the following salts will be neutral, acidic or alkaline. Justify your answers.
   (a) Ammonium nitrate, NH_4NO_3
   (b) Potassium propanoate, CH_3CH_2COOK
   (c) Sodium nitrate, $NaNO_3$
2. What information is required in order to make a prediction about the pH of an aqueous solution of ammonium methanoate, $HCOONH_4$?

SKILLS PROBLEM-SOLVING

3. Calculate K_a for chloroethanoic acid from the following data. 1.89 g of chloroethanoic acid was dissolved in 50 cm^3 of water and the solution was diluted to 250 cm^3 in a volumetric flask. The pH of this solution was 1.99.